

JGH PROJECT INTELLIGENCE ENGINE

PHASE 0 COMPREHENSIVE EXPERT REVIEW SHEET & SPECIFICATION SIGN-OFF

Section 1: Executive Summary & Database Scope

This document presents the complete Phase 0 database introspection report for **jghMasterDB**. Domain experts must review and formally approve: (1) Exhaustive user role definitions, (2) Core schema join paths, (3) Temporal date filters for time-based analytics, (4) Expanded business metrics SQL definitions, and (5) Security blacklists.

Target Database:	jghMasterDB (MySQL 8.0)	Total Tables Analyzed:	238
Profiled Enums:	491 Columns Profiled	Restricted Security Cols:	16 Blacklisted
Candidate Join Edges:	32,251 Graph Edges	Generated Timestamp:	2026-08-12 11:00:07
Review Purpose:	EXHAUSTIVE DOMAIN EXPERT SIGN-OFF	Target Scope:	8 Core Target Tables + Master DB

Section 2: Exhaustive User Role Mapping & Low-Cardinality Enums

Below is the complete inventory of all distinct **user_role** integer codes extracted from users table (29,063 total rows). Domain experts must confirm the exact role name and operational category for EVERY role integer.

Role Code	User Count in DB	% of Users	Proposed Role Name / Title	Operational Category & Expert Sign-Off Line
user_role = 1	6	0.02%	Executive / Staff	Category: Internal Staff [] Approved
user_role = 2	27,703	95.32%	Retailer / Mechanic / End User	Category: Primary Earners [] Approved
user_role = 3	6	0.02%	Area Executive / Sales Rep	Category: Field Team [] Approved
user_role = 4	457	1.57%	Distributor / Channel Partner	Category: Supply Chain [] Approved
user_role = 5	573	1.97%	Wholesaler / Stockist	Category: Supply Chain [] Approved
user_role = 6	248	0.85%	Super Distributor / Enterprise	Category: Supply Chain [] Approved
user_role = 7	1	0.00%	System Admin / Master User	Category: System Control [] Approved
user_role = 8	6	0.02%	Company Auditor / Finance	Category: Operations [] Approved
user_role = 10	23	0.08%	Field Executive	Category: Field Team [] Approved
user_role = 11	10	0.03%	Regional Manager	Category: Management [] Approved
user_role = 13	3	0.01%	Support Manager	Category: Support [] Approved
user_role = 14	1	0.00%	Test / Sandbox User	Category: Testing [] Approved
user_role = NULL	25	0.09%	Unassigned Profile	Category: Legacy Unassigned [] Approved

Target Low-Cardinality Enum Summary

Table & Column Name	Distinct Database Values & Row Counts	Expert Sign-Off & Mapping Rules
wallet_transaction.reference_type	cash_point (6.36M), withdrawal (35.9K), topup (117.5K), incentive (5.0K), bonus_conversion (3.4K), referral_earning (2.6K), credit_note (511)	Earnings: cash_point, topup, incentive, bonus_conversion, referral, credit_note Redemptions: withdrawal [] Approved
wallet_transaction.status	1 (100,000 Sampled / Active Completed Transactions)	Status 1 = Completed / Successful Financial Transaction [] Approved
sku_inventories.order_type	Distributor (52,106), Distributors (47,894)	Wholesaler/Distributor Orders = 'Distributor', 'Distributors' Retailer Orders = 'Retailer' [] Approved
sku_inventories.sources	kolkata (100,000)	Source Warehouse = 'kolkata' (Primary Facility) [] Approved
sku_qr_points_maps.gride_type	FG_BIG (6,724), FG_SMALL (1,250), NULL (90)	FG_BIG = Finished Goods Big Box FG_SMALL = Finished Goods Small Box [] Approved
sku_qr_points_maps.uom	B10 (3,594), B5 (3,067), B6 (64), B12 (22), B3 (1,142), B1 (153), " (22)	B10 = 10 Pcs Box, B5 = 5 Pcs Box, B3 = 3 Pcs Box, B1 = 1 Pc Box [] Approved

Section 3: Core Schema Join Path Map

Below are the verified relational join paths across the 8 target scope tables. These paths define how SQL queries link user profiles, transactions, inventories, payouts, and SKU points.

Source Entity	Join Condition & Keys	Target Entity	Business Purpose / Relationship Context	Expert Sign-Off
users	users.id = wallet_transaction.user_id	wallet_transaction	User financial ledger transactions (credits & debits)	[] Approved
users	users.id = sku_inventories.status_retailer_id	sku_inventories	Retailer scan history & product box assignment	[] Approved
users	users.id = sku_inventories.status_wholeseller_id	sku_inventories	Wholesaler scan history & box movement	[] Approved
users	users.id = sku_inventories.distributor_id	sku_inventories	Distributor inventory allocation & dispatch	[] Approved
sku_inventories	sku_inventories.sku_code = sku_qr_points_maps.sku_code	sku_qr_points_maps	Product UOM box count, cash points, & extra points lookup	[] Approved
users	users.id = withdrawal_request.user_id	withdrawal_request	User cash payout withdrawal requests	[] Approved
withdrawal_request	withdrawal_request.automatic_transaction_id = automatic_transactions.id	automatic_transactions	Bank API payout execution status & bank reference number	[] Approved
companies	companies.business_info_id = users.business_info_id	users	Enterprise business unit & company user grouping	[] Approved

Section 4: Temporal & Date Column Definitions

Standard temporal date columns used for time-filtered queries (e.g. 'July 2026' or monthly reporting). Domain experts must verify these columns are the canonical date filters for time-based metrics.

Table Name	Date Column Name	Data Type	Analytics Usage & Time Filtering Purpose	Verification
wallet_transaction	created_at	TIMESTAMP	Transaction execution timestamp (Standard date filter for monthly Earnings & Redemptions)	[] Verified
sku_inventories	retailer_scanned_at	TIMESTAMP	Retailer QR scan timestamp (Standard filter for Retailer Box Scans in month X)	[] Verified
sku_inventories	wholeseller_scanned_at	TIMESTAMP	Wholesaler QR scan timestamp (Standard filter for Wholesaler Box Scans in month X)	[] Verified
sku_inventories	created_at	TIMESTAMP	Inventory batch generation timestamp	[] Verified
users	created_at	TIMESTAMP	User registration & onboarding timestamp	[] Verified
withdrawal_request	created_at	TIMESTAMP	Payout request initiation timestamp	[] Verified

Section 5: Expanded Business Metrics Catalogue

Below is the complete specification of core business metrics, comparing static profile columns against dynamic ledger expressions. Domain experts must confirm the exact SQL logic and business definitions.

Metric Name	Target Table & Column	Proposed SQL Expression	Business Definition & Expert Sign-Off
1. Static Profile Balance	users.wallet_balance	SUM(wallet_balance)	Current stored wallet cash balance snapshot [] Approved
2. Dynamic Ledger Balance	wallet_transaction.amount	SUM(CASE WHEN reference_type != 'withdrawal' THEN amount ELSE -amount END)	Net calculated ledger balance from financial transactions [] Approved
3. Gross Earnings	wallet_transaction.amount	SUM(amount) WHERE reference_type IN ('cash_point', 'topup', 'incentive', 'bonus_conversion', 'referral_earning', 'credit_note')	Total monetary credits earned by users [] Approved
4. Total Redemptions	wallet_transaction.amount	SUM(amount) WHERE reference_type = 'withdrawal'	Total cash payouts redeemed via withdrawal [] Approved
5. Retailer Box Scans	sku_inventories.order_type	COUNT(*) WHERE order_type = 'Retailer' AND retailer_scanned_at IS NOT NULL	Verified product box scans completed by retailers [] Approved
6. Wholesaler Box Scans	sku_inventories.order_type	COUNT(*) WHERE order_type IN ('Distributor', 'Distributors') AND wholeseller_scanned_at IS NOT NULL	Verified product box scans completed by wholesalers [] Approved
7. Total Points Earned	sku_qr_points_maps.retailer_cash_points	SUM(q.retailer_cash_points + q.wholeseller_cash_points) FROM sku_inventories i JOIN sku_qr_points_maps q ON i.sku_code = q.sku_code	Total reward points generated from box scans [] Approved

Section 6: Restricted & Sensitive Column Security Blacklist

Complete list of 16 detected sensitive columns blacklisted from LLM context prompts and query response outputs.

Table Name	Column Name	Data Type	Keyword Matched	Security Policy & Approval
campaign_logs	device_token	varchar	token	[] Blacklist Approved
company_controls	key	varchar	key	[] Blacklist Approved
connection_request	auth_code	varchar	auth	[] Blacklist Approved
employees	password	varchar	password	[] Blacklist Approved
notification_logs	device_token	varchar	token	[] Blacklist Approved
notifications	device_token	varchar	token	[] Blacklist Approved
password_resets	token	varchar	token	[] Blacklist Approved
personal_access_tokens	tokenable_type	varchar	token	[] Blacklist Approved
personal_access_tokens	tokenable_id	bigint	token	[] Blacklist Approved
personal_access_tokens	token	varchar	token	[] Blacklist Approved
tripur_api_tokens	access_token	text	token	[] Blacklist Approved
tripur_api_tokens	token_type	varchar	token	[] Blacklist Approved
users	password	varchar	password	[] Blacklist Approved
users	two_factor_secret	text	secret	[] Blacklist Approved
users	remember_token	varchar	token	[] Blacklist Approved
users	fcm_token	varchar	token	[] Blacklist Approved

Section 7: Formal Domain Expert Sign-Off & Approval

JGH INTELLIGENCE ENGINE — PHASE 0 FORMAL SIGN-OFF

I hereby confirm that I have reviewed the database profile, user roles, join paths, date filters, business metrics, and security blacklists contained in this document.

☐ Section 2: User Roles Approved

☐ Section 3: Join Paths Approved

☐ Section 4: Date Filters Verified

☐ Section 5: Metrics Approved

☐ Section 6: Security Blacklist Approved

Domain Expert Name: _____

Title / Designation: _____

Signature: _____ Date: _____